Emmanouil Karystinaios



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Technical Skills

- **Programming** : Python, PyTorch, PyG, NumPy, Pandas, Git, Linux
- **Machine Learning** : OMR, Transformers, Graph Neural Networks, Generative Models, Source Separation, DSP
- **ML Engineering** : MLOps, Evaluation, Data Pipelines, CI/CD, Docker, ONNX, Hugging Face
- **Music Tech** : MuseScore, MusicXML, MEI, MIDI, Ableton, Reaper, Max/MSP

Languages

- English
- French
- Greek
- Spanish
- German

Other

- Music performance (Guitar, Trumpet, Piano)
- DAW mixing and production
- Music and film composition

Senior ML Engineer – Music AI

Profile

Senior Machine Learning Engineer at Muse Group, developing robust optical music recognition systems for MuseScore. Music AI researcher with a Ph.D. in Artificial Intelligence and expertise in OMR, symbolic music understanding, graph neural networks, transformers, and audio ML. Published 30+ peer-reviewed papers and built open-source, interactive, and agentic tools that translate research into practical music-technology workflows.

Professional Experience

Muse Group (MuseScore) 2026 – present

- Develop OMR models and evaluation pipelines for converting sheet-music images and PDFs into editable scores.
- Improve architectures, datasets, error metrics, MLOps, and reproducible benchmarking, bridging research with production requirements.

Johannes Kepler University Linz 2020 – present

- Postdoctoral researcher at the Institute of Computational Perception, currently affiliated part-time.
- Research symbolic music analysis, multimodal generation, audio restoration, and explainable music AI; published 30+ peer-reviewed papers.
- Contribute to interdisciplinary research grant proposals with academic and industry partners.

University Teaching 2021 – 2023

- Lecturer in Music and Sound Computing at the University of Music and Performing Arts Vienna and University of Arts Linz.

Selected Projects

Interactive Symbolic Music Analysis 2024 – present

Human-in-the-loop analysis with AnalysisGNN and language models.

- Developed graph-based analysis and interactive workflows for prediction, completion, explanation, and user-guided revision.

ScorePrompts 2026

Interactive, analysis-informed LLM explanations of music scores.

- Created an open score-to-prompt system that turns musical structure into tailored, context-rich explanations; accepted as an ISMIR 2026 Late-Breaking Demo.

MuseScore OMR Production Pipeline 2026 – present

Top-tier optical music recognition for editable scores on MuseScore.

- Develop the production pipeline for converting uploaded scores into editable notation. Try the converter on MuseScore.

Education

Johannes Kepler University Linz 2021 – 2024

Ph.D. in Artificial Intelligence. Thesis: *Symbolic Music Analysis with Graph Neural Networks.*

Diderot University Paris 7 2018 – 2020

M.Sc. in Mathematical Logic and Programming Fundamentals.

Descartes University Paris 8 2017 – 2019

M.A. in Composition and Musical Programming.

Aristotle University of Thessaloniki 2012 – 2017

Diploma in Musicology (Integrated Master's in Composition and Analysis).